QBioS Laboratory: 'Anomalous' Predator-Prey Dynamics

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I. FROM PREDATION EVENTS TO POPULATION DYNAMICS

The study of predator-prey dynamics lies at the very heart of quantitative biosciences. There is now over 100 years of history in efforts to understand how feedback between individual predator-prey events can lead to unexpected dynamics at the population scale. The core ideas are straightforward. Predators consume prey, thereby reducing prey densities and increasing predator densities. In turn, as predators become more abundant and prey less abundant, then predators have, relatively speaking fewer prey to consume. As such, the predator populations begin to decline, reducing predation pressure, and enabling the prey population to increase again. At this point with replete prey, the predators can thrive, and the cycle repeats. This cycle, reviewed in the main text, is known as the canonical predator-prey cycle or the Lotka-Volterra cycle. It has, as its hallmark, a particular ordering of peaks and troughs, that is, first the prey peaks, then the predator peaks, then the prey reaches its lowest point, then the predator, and then prey rebounds again. Yet as helpful as 100 years of history can be, such canonical understanding can also be limiting. If we continue to assume that predator-prey cycles always behave a certain way, then we are less likely to see the data in the way that it should be interpreted, i.e., viewing key biological observations as 'exceptions' rather than as mechanisms worthy of exploration.

What if predator peaks don't always follow prey peaks? What happens if, instead, such predator-prey dynamics are in phase, out of phase, or even 'reversed' in orientation. Indeed, all of these 'anomalous' features do occur in real systems, and not just seldom. The main text treats this issue at length, including the theoretical basis and evidence for non-canonical cycles in natural and experimental systems. The common-place nature of such 'exceptions' arises, in part, because two key assumptions of the canonical predator-prey dynamics models are rarely met (see Figure 1). These violations of assumptions also provide a natural entry point for the laboratory.

The two key assumptions that will be examined by this computational laboratory are:

The speed of evolution Many ecological models assumes that populations are homogeneous, i.e., are comprised of individuals that have the same phenotypes (at least with respect to predator-prey interactions). This assumption makes an implicit assumption about time-scales, i.e., that evolutionary change is very slow compared to ecological dynamics. However, such assumptions are also violated, in that evolution can be fast/rapid such that multiple genotypes of predator and/or prey can coexist at a given moment, such that the total population dynamics of predator and prey behave quite differently than expected.

The form of contact rates : Population dynamic models often make an assumption that contact rates (and indeed clearance rates) of prey by predators is linear, when in fact such interaction rates are 'nonlinear'. These nonlinear interactions arise because of many biological mechanisms, including the time it takes to locate and handle prey.

There are of course many other directions we could choose to explore, from the effect of demographic stochasticity to space. In the first part, the lab focuses on what can happen when more than one type of prey is present in a community, and how projecting a higher dimensional system into a lower dimensional space can help reconcile seemingly impossible violations of the canonical predator-prey dynamics. In the second part, the lab explores the effects of functional response on dynamics, and then goes deeper, exploring the roots of functional responses via an individual based model.

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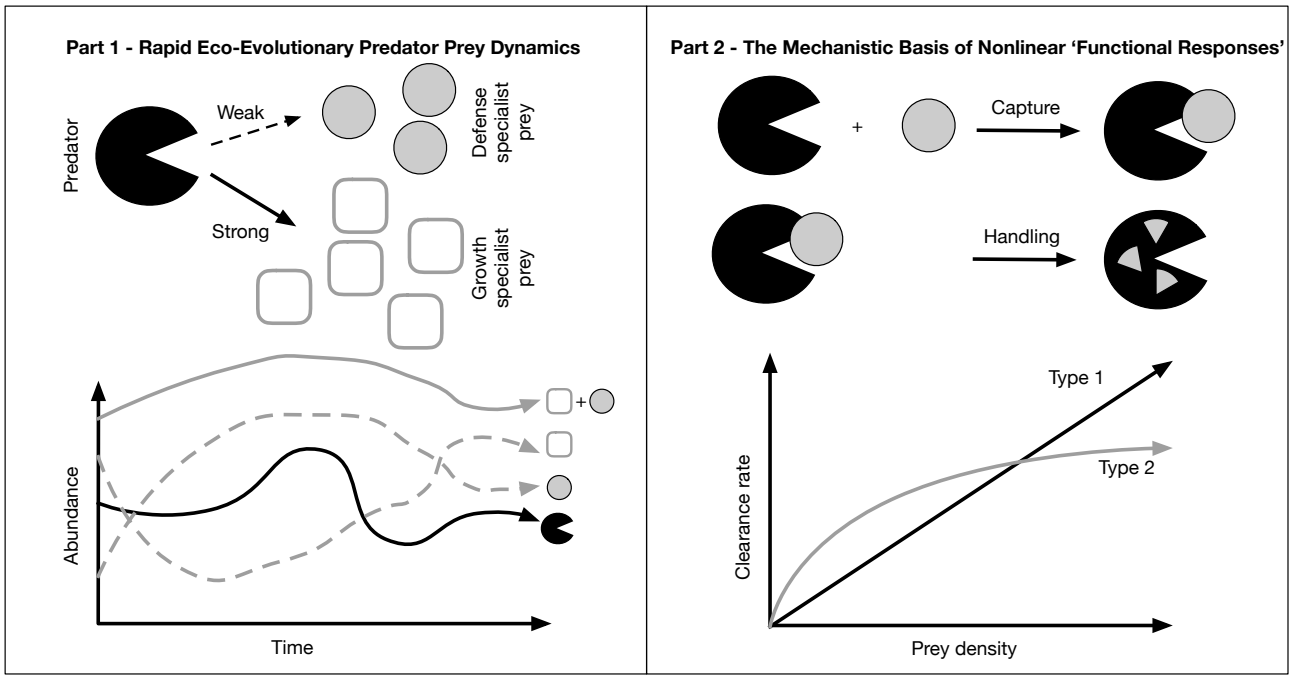


FIG. 1: Organization of this laboratory from population models including ecological and evolutionary dynamics (in Part 1) to the mechanistic basis of functional responses (in Part 2). As will be apparent, variation in the behavior of predators underlies both how many prey can be consumed but how the population and trait dynamics of predators and prey change.

II. ECOLOGICAL DYNAMICS WHEN EVOLUTION IS RAPID/FAST

A. Canonical predator-prey dynamics

The standard predator-prey dynamic model given N prey and P predators with prey-density limitation is:

$$\begin{aligned}\dot{N} &= rN(1-N/K) - bNP \\ \dot{P} &= cNP - mP.\end{aligned}$$

Here r denotes maximum prey growth rate, K is the prey carrying capacity, b is the clearance rate, c is the conversion rate from prey to predator (e.g., $c = \epsilon b$ where $\epsilon < 1$), and m is the predator decay rate in the absence of prey. This code can be separated into two parts, a model function and a script to evaluate dynamics. The model for this system can be written as

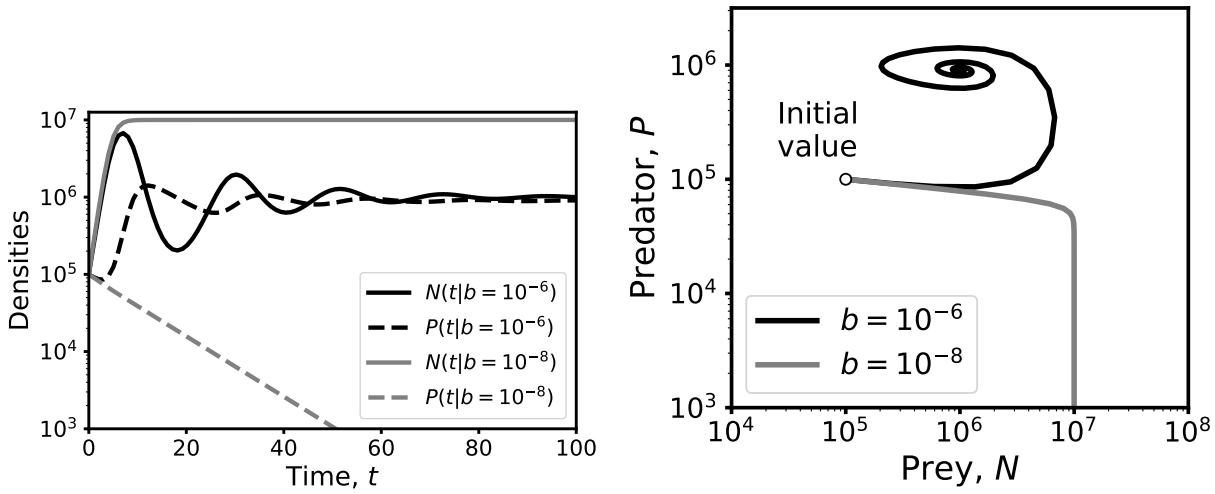
```
def model_pp(y,t,pars):
    """
    Returns the rate of change of prey and predator in the standard LV model
    """
    dydt = np.zeros(2)
    dydt[0] = pars['r']*y[0]*(1-y[0]/pars['K'])-pars['b']*y[0]*y[1]
    dydt[1] = pars['c']*y[0]*y[1]-pars['m']*y[1]
    return dydt
```

These dynamics can be shown to lead to a convergent equilibrium insofar as $N^* < K$ or equivalently $\frac{m}{c} < K$, such that predators can grow on levels of prey that are below the carrying capacity. The derivation of this result is found in the main text, and underlies the first challenge problem.

Challenge Problem: Convergent Predator-Prey Dynamics

Consider zooplankton (microscopic predators) consuming bacteria (their prey) such that $r = 1$ hrs^{-1}, $K = 10^7$/ml, $c = 0.1b$, and $m = 0.1$ hrs^{-1}. Find the critical value of the clearance rate b that enables predator-prey coexistence, and show the results of dynamics as functions of time or in the phase plane (i.e., prey on the x-axis and predators on the y-axis).

Solutions to Challenge Problem: Convergent Predator-Prey Dynamics:
Note that predator-prey coexistence requires that $1 - N^*/K > 0$ or that $N^* < K$. Because prey densities are controlled by predators, $N^* = m/c$ which in this example is $N^* = 10m/b$, such that the condition on the clearance rate enabling predators to survive on prey is: $b > \frac{10m}{K}$ or $b > 10^{-7}$ (ml-hr)$^{-1}$. Hence, the following figures show starkly different outcomes given choices of b, either 10-fold above and below that critical value.



The phase plane figures are generated using the following code (legend optional):

```
# Units of hrs, ml
pars={}
pars['r']=1
pars['K']=10**7
pars['eps']=0.1
pars['m']=0.1
pars['T']=100

fig = plt.figure()
ax = fig.gca()

# Coexistence case
pars['b']=10**-6
pars['c'] = 0.1*pars['b']
t =np.linspace(0,pars['T'],100)
y0=10**5*np.ones(2)
y = integrate.odeint(model_pp,y0,t,args=(pars,))
plt.loglog(y[:,0],y[:,1],'k',linewidth=3)

# Failure case
pars['b']=10**-8
pars['c']=0.1*pars['b']
y = integrate.odeint(model_pp,y0,t,args=(pars,))
plt.loglog(y[:,0],y[:,1],color=[0.5,0.5,0.5],linewidth=3)

plt.loglog(y0[0],y0[1],marker='o',markeredgecolor='k',markerfacecolor='w')

plt.xlim(10**4,10**8)
plt.ylim(10**3,10**6.5)
plt.text(10**4.4,10**5.2,'Initial\nvalue',fontsize=16)
```

B. Functional responses – a phenomenological view

The standard predator-prey dynamic model given N prey and P predators with prey-density limitation and a generalized 'functional response' is:

$$\begin{aligned}\dot{N} &= rN(1-N/K) - bf(N)P \\ \dot{P} &= cf(N)P - mP.\end{aligned}$$

where $f(N)$ denotes the functional response, i.e., denoting how the clearance rate of prey by predators scales with prey density. The standard approach to classify such responses is

$$\text{Type1}: f(N) \sim N \tag{1}$$

$$\text{Type2}: f(N) \sim \frac{N}{Q+N} \tag{2}$$

$$\text{Type3}: f(N) \sim \frac{N^2}{Q^2+N^2} \tag{3}$$

These functional response modify the stability of the coexistence fixed point (see chapter notes for more details). As a means to explore this, consider a Type 2 functional response in a particular predator-prey system:

$$\dot{N} = rN(1-N/K) - \frac{bN}{1+b\tau N}P \tag{4}$$

$$\dot{P} = \frac{\epsilon bN}{1+b\tau N}P - mP, \tag{5}$$

where τ is an effective handling time. This is known as the MacArthur-Rosenzweig model. A modified code to implement this model is as follows:

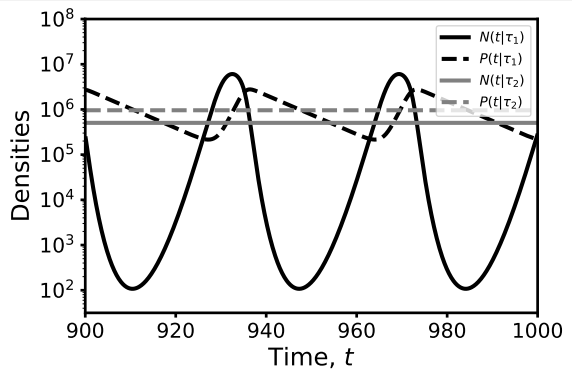
```
def pp_dyn_type2(y,t,pars):
    """
    predator-prey dynamics
    """
    N=y[0]
    P=y[1]

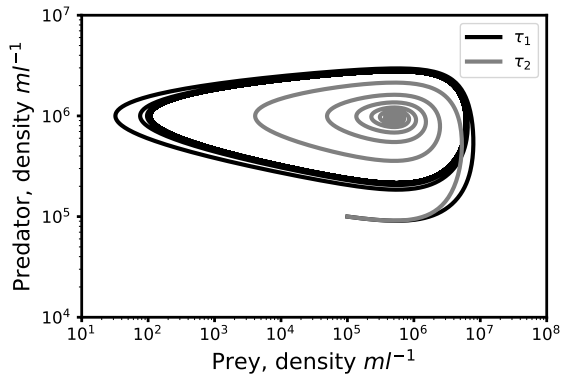
    dydt = np.zeros(2)
    dydt[0] = pars['r']*N*(1-N/pars['K'])-pars['b']*N*P/(1+pars['b']*pars['tau']*N)
    dydt[1] = pars['eps']*pars['b']*N*P/(1+pars['b']*pars['tau']*N)-pars['m']*P

    return dydt
```

Challenge Problem: From Convergence to Limit Cycles

Consider zooplankton (microscopic predators) consuming bacteria (their prey) such that $r = 1$ hrs^{-1}, $K = 10^7$/ml, $b = 10^{-6}$ (ml-hr)$^{-1}$, $\epsilon = 0.2$, and $m = 0.1$ hrs^{-1}. Try to find a value of τ that yields limit cycle dynamics and then, if possible, modify τ to change dynamics so that they only exhibit convergent dynamics to a fixed point. If it works, your output should look like the following image where the values of τ_1 and τ_2 are yours to discover!





Solutions to Challenge Problem: From Convergence to Limit Cycles:
The key to this challenge problem is to recognize that τ should have units similar to that of the growth rate. The detailed examination of the coexistence criteria and transition to limit cycles are examined in the main text. Here, these examples are for $\tau_1 = 0.2$ (limit cycles) and $\tau_2 = 0.02$ (convergence to a fixed point). It is worth noting that in the limit as $\tau \to 0$, the Type 2 functional response converges to a Type 1 functional response where limit cycles are not expected. The code for these simulations is similar to that above, modified given the invocation of different model functions. Hence, as an alternative, here is the code for the time-series view, focusing in only on the last 100 hrs of the simulation:

```
t=np.arange(0,1000,0.1)
y0=10**5*np.ones(2)

fig=plt.figure()
ax = fig.gca()
#Limit cycle case
pars['tau']=0.2
y = integrate.odeint(pp_dyn_type2,y0,t,args=(pars,))
tmpi = np.where(t>900)
plt.plot(t[tmpi],np.transpose(y[tmpi,0]),'k',linewidth=3)
plt.plot(t[tmpi],np.transpose(y[tmpi,1]),'k',linestyle='--',linewidth=3)

#no limit cycles
pars['tau']=0.02
y = integrate.odeint(pp_dyn_type2,y0,t,args=(pars,))
tmpi = np.where(t>900)
plt.plot(t[tmpi],np.transpose(y[tmpi,0]),color=[0.5,0.5,0.5],linewidth=3)
plt.plot(t[tmpi],np.transpose(y[tmpi,1]),color=[0.5,0.5,0.5],linewidth=3,linestyle='--')

ax.set_yscale('log')
```

C. Steps towards rapid eco-evolutionary dynamics

The prior model assumed that interactions take place between a homogeneous population of prey and predators. However what happens if there are, in fact, two distinct types of prey, one that is a growth-specialist and one that is a defense-specialist. It is often the case that such latent variation exists in the environment, even if it is not always

possible to differentiate prey (or predators) in this way. The reasons are varied, including the possibility that genetic analysis is unfeasible, counts of types are based on morphological features (for which there is cryptic variation), and that sites of variation may be unrelated to the predator-prey interaction. For example, if one were to use 16S rRNA metrics of diversity, then many bacteria would appear the same, even if their functional properties were radically different.

Hence, consider a variation of the standard predator-prey model in which there are two types of prey, N_1 and N_2 interacting with one predator:

$$\frac{\mathrm{dN_1}}{\mathrm{dt}} = r_1 N_1 \left(1 - \frac{N_1 + a_{12} N_2}{K_1}\right) - \frac{c_1 N_1 P}{(1 + c_1 N_1 + c_2 N_2)} \tag{6}$$

$$\frac{\mathrm{dN_2}}{\mathrm{dt}} = r_2 N_2 \left(1 - \frac{N_2 + a_{21} N_1}{K_2}\right) - \frac{c_2 N_2 P}{(1 + c_1 N_1 + c_2 N_2)} \tag{7}$$

$$\frac{\mathrm{dP}}{\mathrm{dt}} = \left(\frac{\epsilon c_1 N_1 + \epsilon c_2 N_2}{1 + c_1 N_1 + c_2 N_2}\right) P - mP \tag{8}$$

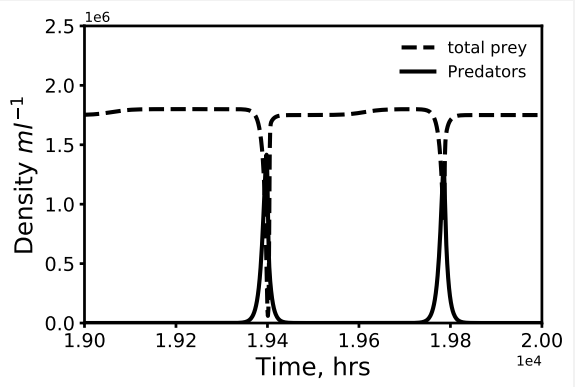
This is a natural extension of the single model, however, there is an important caveat. We may be interested in the case of dynamics for total prey and total predator, i.e., in the $N - P$ plane where $N \equiv N_1 + N_2$. Hence, below is a code base that implements a variant of this model. If working, it should reveal **anti-phase** dynamics, where predators seem to peak at precisely the moment (more or less) when prey are reaching their trough. The reason for this is complicated, and involves rapid switches in the relative frequency of prey genotypes, i.e., rapid evolution on ecological time-scales! But, some of that must be left for the homework. You may want to extend and modify this code to see the shift in genotypes for yourself

Challenge Problem: Eco-Evolutionary Dynamics

Write the model described in equations 6 and run it with the following parameters:

```
#Parameters
pars={}
pars['r1']=1.8 #hr^-1
pars['r2']=1.75 #hr^-1
pars['c1']=3 # ml/hr
pars['c2']=0.3 # ml/hr
pars['tau'] = 1
pars['K1']=1.8 # cells/ml
pars['K2']=1.75 # cells/ml
pars['eps']=0.5 # conversion
pars['m']=0.3 #hr^-1
```

The resulting figure should look like that below, and note that a 'stiff' integration method may help accelerate the simulations:



Solutions to Challenge Problem: Eco-Evolutionary Dynamics (part 1 of 2):
The model described in equations 6 can be written as follows:

```
def pp_dyn_evo(y,t,pars):
    N1 = y[0]
    N2 = y[1]
    P = y[2]

    N = N1+N2

    dydt = np.zeros(3)
    Q = pars['c1']*N1 + pars['c2']*N2
    dydt[0] = pars['r1']*N1*(1-N/pars['K1'])-pars['c1']*N1*P/(1+Q)
    dydt[1] = pars['r2']*N2*(1-N/pars['K2'])-pars['c2']*N2*P/(1+Q)
    dydt[2] = pars['eps']*(Q/(1+Q))*P-pars['m']*P

    return dydt
```

Solutions to Challenge Problem: Eco-Evolutionary Dynamics (part 2 of 2):
The following is the master script to run the full model of eco-evolutionary dynamics.

```
#Parameters
pars={}
pars['r1']=1.8 #hr^-1
pars['r2']=1.75 #hr^-1
pars['c1']=3 # ml/hr
pars['c2']=0.3 # ml/hr
pars['tau'] = 1
pars['K1']=1.8 # cells/ml
pars['K2']=1.75 # cells/ml
pars['eps']=0.5 # conversion
pars['m']=0.3 #hr^-1

sfactor = 10**6

fig=plt.figure()
ax = fig.gca()
#model and visualization
t = np.arange(20000)
y0=np.array([0.13,1.35,1.2])
y = integrate.odeint(pp_dyn_evo,y0,t,args=(pars,),rtol=10**-7,hmax=1)
N = y[:,0]+y[:,1]
plt.plot(t,N*sfactor,'k',linewidth=3,linestyle='--')
plt.plot(t,y[:,2]*sfactor,'k',linewidth=3)

plt.ylim(0,2.5*sfactor)
plt.xlim(19000, 20000)
plt.xlabel(r'Time, hrs',fontsize=18)
plt.ylabel(r'Density $ml^{-1}$',fontsize=18)

plt.legend(['total prey','Predators'],fontsize=12,loc='upper right',
           frameon=False)
ax.ticklabel_format(style='scientific',scilimits=(0,0))
plt.setp(ax.spines.values(),linewidth=2)
ax.tick_params(labelsize=14,width=2)
```

III. FUNCTIONAL RESPONSES - A MICROSCOPIC APPROACH

A. Theoretical background

Predators or consumers must interact with their prey or resource. Here, we will consider the "microscopic" basis for the kinds of functional forms used to described predator-prey and consumer-resource models. Recall that the Type 1 predator-prey model equations are:

$$\begin{aligned}\dot{N} &= rN(1-N/K) - bNP \\ \dot{P} &= cNP - mP\end{aligned}$$

where the term bNP denotes consumption of prey by predators. This form suggests a collision-based approach, but does it also apply to spatial dynamics? Moreover, how would this kind of functional form changed if predators had to "handle" hosts before consuming them. This lab and the homework will help you understand these questions.

First, let's assume that a predator moves "ballistically" through an environment with a velocity v. Moreover, assume the predator can sense prey in a zone of radius r. Hence, the predator will sense an area of approximate size $\pi(r+vT_s)^2$ in a period T_s of searching. If prey are at a density N, then the potential number of prey encounters is $(\pi r^2 + 2rvT_s)N$. Further if a fraction k are detected and of those f are consumed per unit time, then the total prey consumption is $\pi(r+vT_s)^2 fkT_sN$ per predator in a period of time T_s. This is true for all predators, as long as they tend *not* to run into or near each other. Hence, we assume that the total consumption per unit time scales like

$$\text{consumption rate} = bNP$$

where the constant b approaches

$$\pi r^2 fk$$

in the limit that $T_s \to 0$. For example, you may find that this limit poses some problems, e.g., an immobile predator with relatively slow diffusing prey may rapidly deplete local prey. So even though it continues to sense across a radius of r, there may not be any prey left to sense! To test these ideas, we will use an individual-based model (IBM) of predator detecting and eating prey. This code should be downloaded first, given its complexity.

B. Getting Started

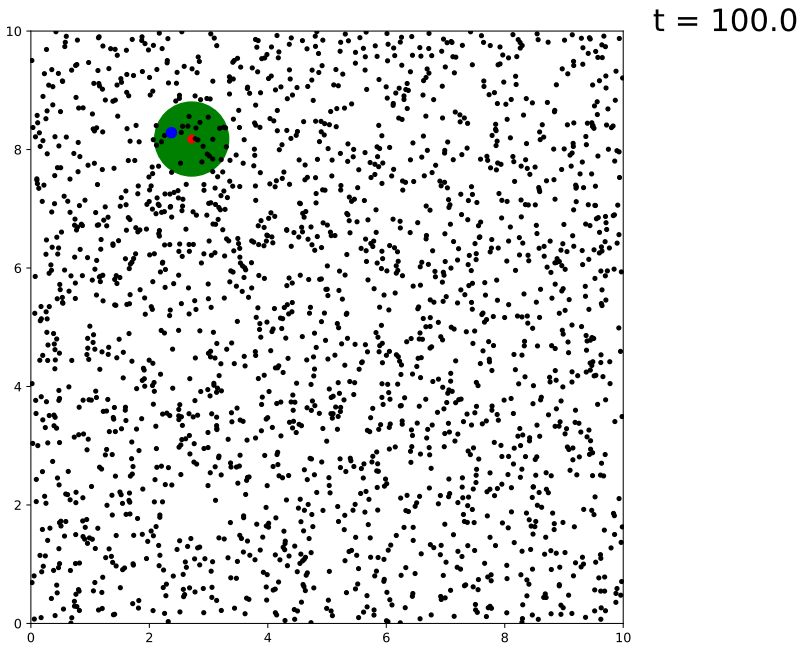
First, download three files:

- `master_ibm.py` – the master script for setting parameters of the IBM model
- `ibm_functions.py` – dynamic simulator of predation as well as utility functions

Once you've downloaded the files, go to Spyder and at the prompt type:

```
%matplotlib qt
```

so that the visualization will display in its own window. Then run the master script. You should see a predator with a zone of detection eating prey. The predator moves ballistically (in a straight line) and changes directions on occasion, whereas the prey move diffusively. Circles that appear denote the last set of prey eaten in a particular interval. These circles automatically update as new prey are consumed. Re-size the window so you can see the update label. This is what it should look like when done:



When the simulation is done, type:

```
>> figure; plot(t,total_eaten);
```

which will give you a display of the cumulative number of prey eaten by the predator with time. Now that the code is working, let's take a closer look:

```
import numpy as np
import ibm_functions

# Basic information
info={}
info['prey_density']=10 # density/cm^2
info['maxX']=10 # cm
info['maxY']=10 # cm
info['tf']=50 # sec
info['dt']=0.1 #time step
info['replenish_prey']=1 #should prey regenerate?
info['viz_dyn']=1 #1 for animation, 0 for no animation

# Define the prey
prey = {}
prey['num'] = info['prey_density']*info['maxX']*info['maxY']
prey['pos']=np.random.uniform(size=(prey['num'],2))
prey['pos'][:,0]=prey['pos'][:,0]*info['maxX']
prey['pos'][:,1]=prey['pos'][:,1]*info['maxY']
prey['diffusion']=0.005 #cm^2/sec

# Place the predator
predator={}
predator['pos'] = np.array( [[ info['maxX']/2, info['maxY']/2 ]] )
predator['theta']=np.random.uniform()*2*np.pi #Angle of movement
predator['r']=1.25 #Radius
predator['k']=0.3 #Detection
```

```
predator['f']=0.05 #successful capture per time
predator['vel']=0.1 #cm/sec
predator['handling_time']=0.0 #sec
predator['tau']=6 #run time length, sec
predator['trun'] = 0 # Initialize each run

# Simulate eating
reps=1
totend = np.zeros(shape=(1,reps))
for ens in np.arange(reps):
    print(ens)
    [t,numeaten] = ibm_functions.ibm_predation(info,predator,prey)
    total_eaten = np.cumsum(numeaten)
```

Do you understand what each of the terms mean? It will be worthwhile to spend some time looking over the somewhat difficult function: ibm_predation(info,predator,prey).

C. Exploration of predation and functional responses

The following are your objectives for the remainder of this section of the lab. Each is a challenge... with solutions to be revealed along the way.

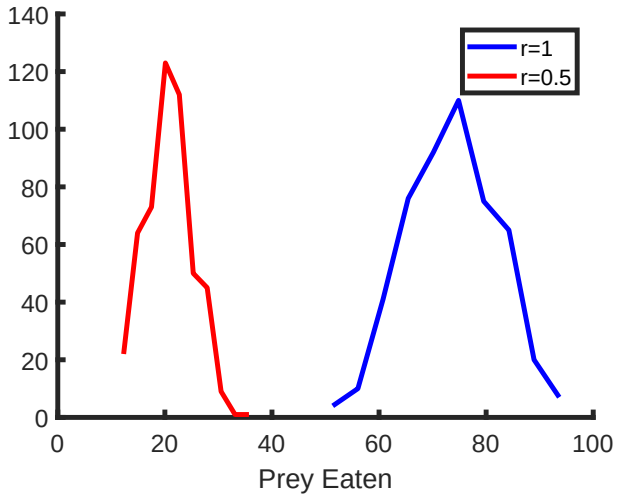
Challenge Problem: Individual-Based Model Zone

Test the hypothesis that reductions in the zone of detection decrease the number of prey eaten by a squared factor. For example, set the predator radius equal to 1 and then while keeping all other parameters the same reduce the predator radius by half and see if there is a 4:1 ratio in the number of prey eaten.

Solutions to Challenge Problem: Individual-Based Model Zone

We can run the model 500 times with predator.r = 1 and predator.r = 0.5 with the following additional parameters to get a histogram

```
info['prey_density']=20 # density/cm^2
prey['diffusion']=0.005 #cm^2/sec
predator['k']=0.3 #Detection
predator['f']=0.05 #successful capture per time
predator['vel']=.1 #cm/sec
predator['handling_time']=0.0 #sec
predator['tau']=6 #run time length, sec
```



It does appear that the mean prey eaten when the detection radius is 1 cm is about 80, while the mean prey eaten when the detection radius is 0.5 cm is about 20.

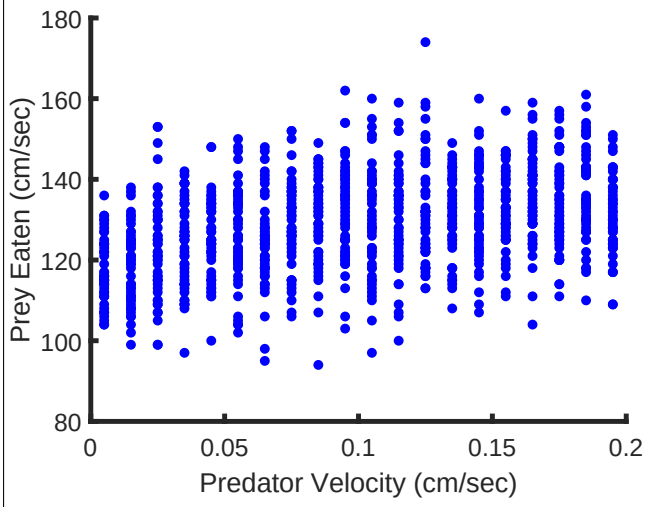
Challenge Problem: IBM Velocity

Evaluate how changes in velocity modify the scaling relationship.

Solutions to Challenge Problem: IBM Velocity.
The velocity of the predator can be modified in the master script. A scatter plot of the number of prey eaten versus the velocity of the predator with the following parameters gives a nearly unchanged relationship over the zone considered.

```
info['prey_density']=20 # density/cm^2
predator['r']=1.25 # Radius
predator['k']=0.3 #Detection
predator['f']=0.05 #successful capture per time
predator['handling_time']=0.0 #sec
predator['tau']=6 #run time length, sec

predator['vel']=np.arange(0.005,0.2,0.01)
```



The number of prey eaten does not change as the velocity changes. To check, we can perform a linear regression of the logs of our data. Doing so gives a slope of 0.048, with a standard error of ± 0.002, close to zero, suggesting that the number of prey eaten is independent of the predator's velocity.

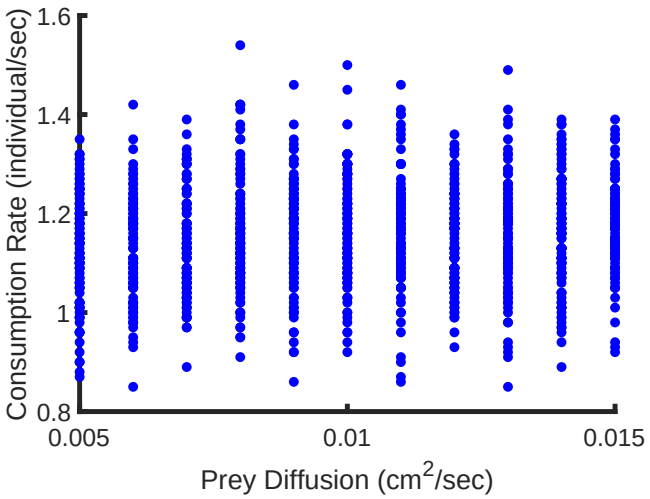
Challenge Problem: IBM Diffusion

Modify the diffusion coefficient of prey, does it modify the consumption rate of predators?

Solutions to Challenge Problem: IBM Diffusion
Vary the diffusion over 1000 repetitions with the following parameters:

```
info['prey_density']=20 # density/cm^2
predator['r']=1.25 # Radius
predator['k']=0.3 #Detection
predator['f']=0.05 #successful capture per time
predator['vel']=0.1 #cm/sec
predator['handling_time']=0.0 #sec
predator['tau']=6 #run time length, sec

prey['diffusion']=np.arange(0.005,0.015,0.001)
```



This yields the following outcome:
The number of prey eaten does not change as the diffusion rate varies, hence this analysis suggests that number of prey eaten is independent of the prey's diffusion rate.

Challenge Problem IBM Density

Change the density of prey (don't increase it beyond 50 or so) - how does prey consumption change.

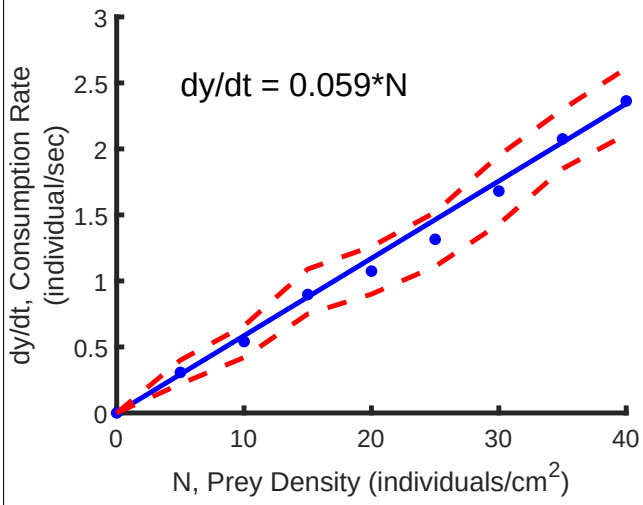
Solutions to Challenge Problem: IBM Density

Consider the variation of the consumption rate given changes in density from 5 to 40 individuals/cm^2. We run the simulation with the following parameters

```
prey['diffusion']=0.005
predator['r']=1.25 # Radius
predator['k']=0.3 #Detection
predator['f']=0.05 #successful capture per time
predator['vel']=0.1 #cm/sec
predator['handling_time']=0.0 #sec
predator['tau']=6 #run time length, sec

info['prey_density']=np.arange(5,40,5) # density/cm^2
```

As below, this is well-fit by a line in log-log space with a slope of 0.99 (i.e., this is a line). Note that the pre-factor is 0.059≈ 0.06, whereas the theory predicts a prefactor of 0.07, so there is near quantitative agreement here.



D. Lions and tigers and handling time, oh my!

Of course, predators do not instantaneously convert prey biomass into predator biomass (or even into new offspring). In the absence of an age structured model with quotas, it is important to at least consider the effect of handling time. Here, it is important to restore all parameters to their original values and set

```
predator['handling_time']=0.5
```

This will make the predator wait 0.5 seconds for each prey it consumes. The model will continue to show which prey has been eaten (via a blue circle), even as the predator moves on. How does the total number of prey consumed compare to the prior case? Can you set parameters such that predators are essentially handling-time limited (hint: set detection and fraction consumed to be nearly 1)? In which case, does your model agree with predictions?

In the homework you will explore how handling time changes the functional response. In particular, when predators must take time to handle and consume their prey then the consumption rate is hypothesized to scale like

$$\text{consumption rate} = \frac{bNP}{1+aN}$$

where b/a is equal to $1/T_h$ where T_h is the handling time. You will get a chance to use this individual-based model to see if and how this hypothesis is valid.